Section 4.2: Measures of Central Tendency

Finite Mathematics MTH 107 Fall 2014

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/home/jim/JamesAshe/MHU/images/MarsHil

Happy Halloween

Data:

$$70, 70, 80, 90, 100$$

Data:

$$70, 70, 80, 90, 100$$

Mean

"average"

Happy Halloween

Data:

$$70, 70, 80, 90, 100$$

Mean

''average''

$\frac{70+70+80+90+100}{5}$

Data:

$$70, 70, 80, 90, 100$$

Mean

"average"

$$\frac{70+70+80+90+100}{5}$$

Median

"middle"

Data:

$$70, 70, 80, 90, 100$$

Mean

"average"

$$\frac{70+70+80+90+100}{5}$$

Median

"middle"

$$70, 70, \boxed{80}, 90, 100$$

Data:

$$70, 70, 80, 90, 100$$

Mean

"average"

$\frac{70+70+80+90+100}{5}$

Median

"middle"

$70, 70, \boxed{80}, 90, 100$

Mode

"most frequent"

Data:

$$70, 70, 80, 90, 100$$

Mean

"average"

$$\frac{70+70+80+90+100}{5}$$

Median

"middle"

$$70, 70, \boxed{80}, 90, 100$$

Mode

"most frequent"

$$\boxed{70,70}, 80, 90, 100$$

The Mean

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Mean Given n data points, $x_1, x_2, \ldots, x_n$, the *mean* is $\bar{x} = \dfrac{\sum x}{n}$.

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$$n = 6$$

$$\begin{aligned} \sum x &= 3 + 15 + 22 + 9 + 17 + 13 \\ &= 79 \end{aligned}$$

Year	TDs
1985	3
1986	15
1987	22
1988	9
1989	17
1990	13

Mean Given n data points, $x_1, x_2, \ldots, x_n$, the *mean* is $\bar{x} = \frac{\sum x}{n}$.

Example 4.2.1.

Find the mean of the given data set.

$$n = 6$$

$$\sum x = 3 + 15 + 22 + 9 + 17 + 13$$
$$= 79$$

$$\bar{x} = \frac{\sum x}{n} = \frac{79}{6} \approx 13.167$$

Year	TDs
1985	3
1986	15
1987	22
1988	9
1989	17
1990	13

Mean: Grouped Data

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	Wage	Freq.	
	$16 \leq x < 20$	$108,000$	
	$20 \leq x < 25$	$53,000$	
	$25 \leq x < 35$	$41,000$	
	$35 \leq x < 45$	$23,000$	
	$45 \leq x < 50$	$29,000$	
	$50 \leq x < 55$	$23,000$	
		$n = 277,00$	

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Example 4.2.2.

Find the mean of the given data set.

	Wage	Freq.	
	$16 \leq x < 20$	$108,000$	
	$20 \leq x < 25$	$53,000$	
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Mean: Grouped Data

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$\sum x = ?$

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$\sum x = ?$

Use the midpoint, $\frac{\text{bottom}+\text{top}}{2}$.

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ex. $\frac{16+20}{2} = 18$

Example 4.2.2.

Find the mean of the given data set.

Midpoint	Wage	Freq.	
18	$16 \leq x < 20$	108,000	
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ex. $108,000 \cdot 18 = 1,944,000$

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Find the mean of the given data set.

Midpoint	Wage	Freq.	$f \cdot x$
18	$16 \leq x < 20$	108,000	1,944,000
22.5	$20 \leq x < 25$	53,000	1,192,500
30	$25 \leq x < 35$	41,000	1,230,000
40	$35 \leq x < 45$	23,000	920,000
50	$45 \leq x < 50$	29,000	1,450,000
60	$50 \leq x < 55$	23,000	1,380,000
		$n = 277,00$	

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$\sum x = ?$

Use the midpoint, $\frac{\text{bottom}+\text{top}}{2}$.

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		$n = 277,00$	$\sum(f \cdot x) = 8,116,500$

Mean: Grouped Data

$n = 277,000$

$\sum x = ?$

Use the midpoint, $\frac{\text{bottom}+\text{top}}{2}$.

ex. $\frac{16+20}{2} = 18$

also need to adjust for the frequency.

ex. $108,000 \cdot 18 = 1,944,000$

Mean:
$\bar{x} = \frac{\sum(f \cdot x)}{n}$

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Use the midpoint, $\frac{\text{bottom}+\text{top}}{2}$.

ex. $\frac{16+20}{2} = 18$

also need to adjust for the frequency.

ex. $108,000 \cdot 18 = 1,944,000$

Mean:
$\bar{x} = \frac{\sum(f \cdot x)}{n} = \frac{8,116,500}{277,00}$

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		$n = 277,00$	$\sum(f \cdot x) = 8,116,500$

The Median

a. 2,4,6,$\boxed{7}$,8,10,12

b. When n is even, take the average of the middle:

Example 4.2.3.

Find the median of the following data:

a. 2,4,6,$\boxed{7}$,8,10,12
b. When n is even, take the average of the middle:

$$2, 4, \boxed{6,8}, 12, 10$$

The Median

Location of Median

Given n data points, the location of the median is $L = \dfrac{n+1}{2}$ after the data has been arranged in ascending order.

Example 4.2.3.

Find the median of the following data:

a. $2, 4, 6, 8, 12, 10, 7$

a. 2,4,6,$\boxed{7}$,8,10,12

b. When n is even, take the average of the middle:

$$2, 4, \boxed{6,8}, 12, 10$$

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a. 2,4,6,$\boxed{7}$,8,10,12

b. When n is even, take the average of the middle:

$2, 4,\boxed{6,8}, 12, 10$

Location: $L = \dfrac{6+1}{2}$ =3.5

data pt between 6 and 8,

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b. When n is even, take the average of the middle:

$2, 4, \boxed{6,8}, 12, 10$

Location: $L = \dfrac{6+1}{2}$ =3.5

data pt between 6 and 8, $\frac{6+8}{2} = 7$

The Mode

The Mode

a. 1 4 5 5 8 10 10 10

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a. 1 4 5 5 8 10 10 10

The mode is 10

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a. 1 4 5 5 8 10 10 10

The mode is 10

b. 1 1 4 4 9 9 10 10

Each data point has the same frequency

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a. 1 4 5 5 8 10 10 10

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Each data point has the same frequency
-there is no mode.

The Mode

a. 1 4 5 5 8 10 10 10

The mode is 10

b. 1 1 4 4 9 9 10 10

Each data point has the same frequency
-there is no mode.

c. 1 3 3 6 8 9 9

$2 \times 3's$ and $2 \times 9's$
The mode is 3 and 9
called *bimodal.*

The Mode

a. 1 4 5 5 8 10 10 10

The mode is 10

b. 1 1 4 4 9 9 10 10

Each data point has the same frequency
-there is no mode.

c. 1 3 3 6 8 9 9

$2 \times 3's$ and $2 \times 9's$
The mode is 3 and 9
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Example 4.2.4.

Find the mode of the following data sets:

The Mode

a. 1 4 5 5 8 10 10 10

The mode is 10

b. 1 1 4 4 9 9 10 10

Each data point has the same frequency
-there is no mode.

c. 1 3 3 6 8 9 9

$2 \times 3's$ and $2 \times 9's$
The mode is 3 and 9
called *bimodal.*

Example 4.2.4.

Find the mode of the following data sets:

a. 4 10 1 8 5 10 5 10

The Mode

a. 1 4 5 5 8 10 10 10

The mode is 10

b. 1 1 4 4 9 9 10 10

Each data point has the same frequency
-there is no mode.

c. 1 3 3 6 8 9 9

$2 \times 3's$ and $2 \times 9's$
The mode is 3 and 9
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Example 4.2.4.

Find the mode of the following data sets:

a. 4 10 1 8 5 10 5 10

b. 4 9 1 10 1 10 4 9

The Mode

a. 1 4 5 5 8 10 10 10

The mode is 10

b. 1 1 4 4 9 9 10 10

Each data point has the same frequency
-there is no mode.

c. 1 3 3 6 8 9 9

$2 \times 3's$ and $2 \times 9's$
The mode is 3 and 9
called *bimodal.*

Example 4.2.4.

Find the mode of the following data sets:

a. 4 10 1 8 5 10 5 10
b. 4 9 1 10 1 10 4 9
c. 9 6 1 8 3 10 3 9